

MicroPythonOS

NewLine – Saturday 30-5-2026

whoami



- Quasi Kili
- Maker and Artist
- Contributor
MicroPythonOS
- Fan of MPOS

whoishe



- Thomas Farstrike
- Creator of MPOS
- Background:
 - Kernel development
 - Embedded Linux
 - C/C++

Origins



- Almost 2 Years Ago
- Building new device
- Need for modern ESP32 Touch OS
- ...with AppStore
- Didn't exist!

Create it in C/C++ ?



- Known territory
- Most ESP32 libraries
- Lightweight
- Logical choice
- Why not?

Create it in C/C++ ?



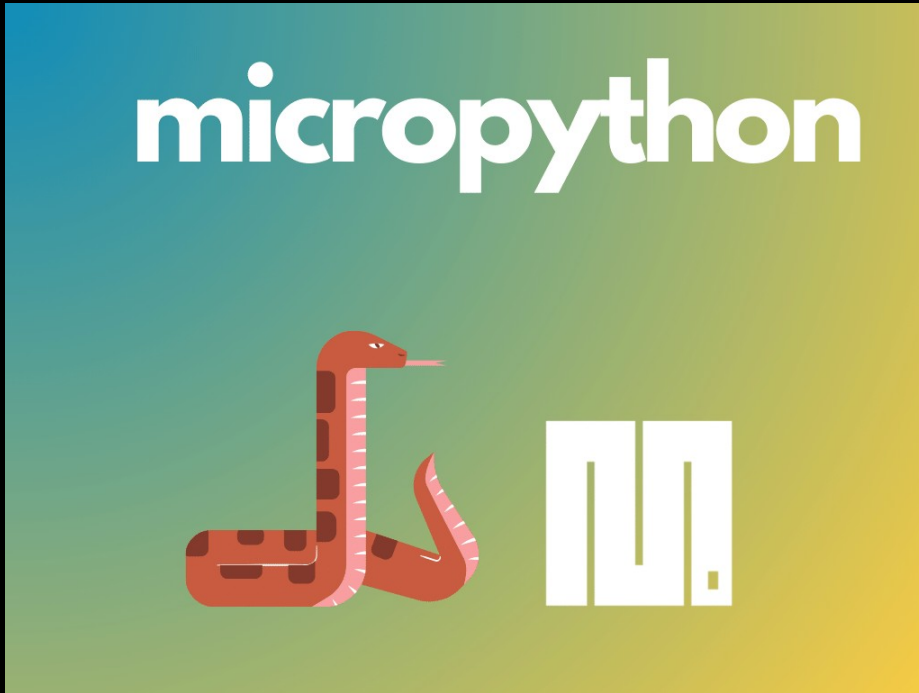
- Limited MMU
- Custom compilation
- Challenge to maintain
- Unforgiving
- Difficult to create OS
- Not easy for app developers

Create it in MicroPython!



- Dynamic loading
- Scripts (.py)
- Binaries (.mpy)
- Modern language
- Easy for app developers
- Drivers in MicroPython

MicroPython



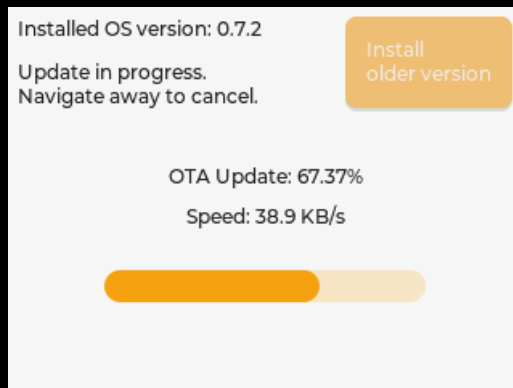
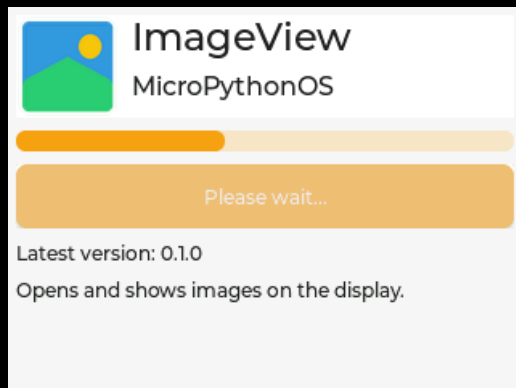
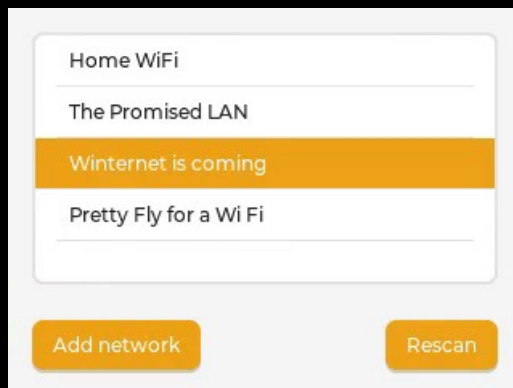
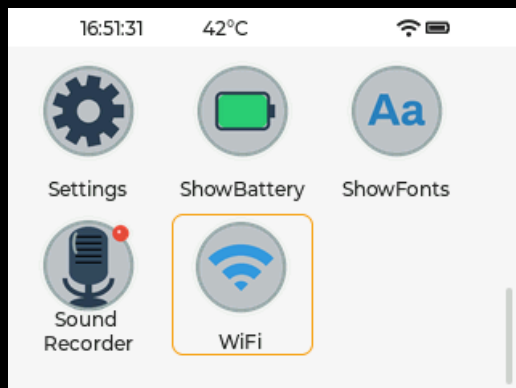
- Born 2014, ESP 2017
- Surprisingly fast
- Many libraries
- Also desktop (native)
- Quick Coding Cycle
- 99% like Python

LVGL makes it beautiful



- Light and Versatile
 - 30+ Widgets
 - Animations
 - Image Decoders
 - UI Editor
 - MicroPython bindings
- Credit:
lvgl_micropython

Proving the concept



- Launcher
- WiFi + Auto-Connect
- AppStore
- OSUpdate
- Settings
- About

High-level architecture



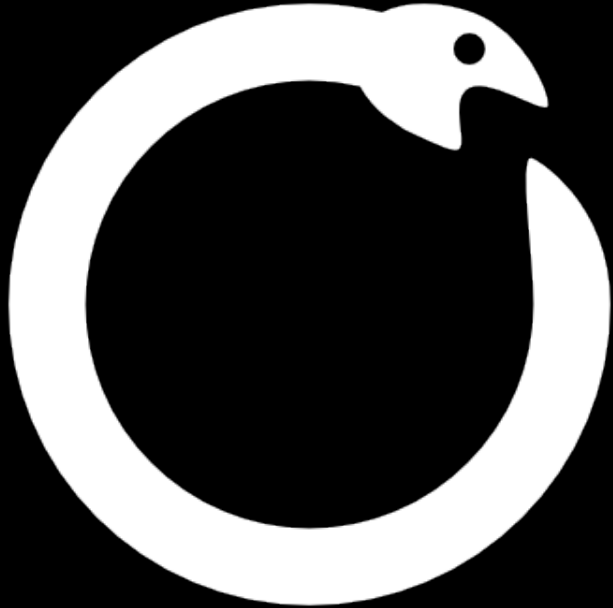
- Apps
- Frameworks:
MicroPython and C
- High-level drivers:
MicroPython and C
- MicroPython Runtime
- Low-level drivers:
FreeRTOS, Darwin, Linux

Lessons from Android



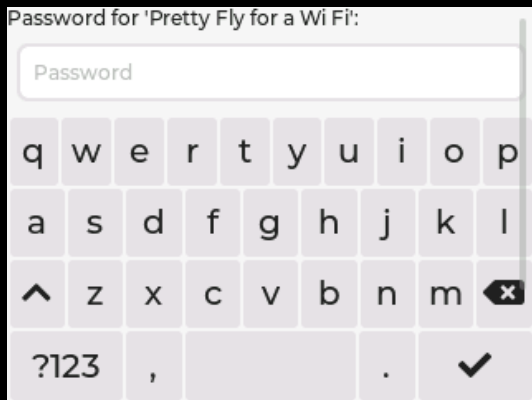
- Create once, run on all (compatible) devices
- Easy app development
- User Interface
- BUT more open source!

Why MicroPythonOS



- Very Lightweight
- Boots in 3 seconds
- Python language
- 100% open development
- Frequent releases
- Runs native on desktop
- Up-and-coming

Innovations



- MposKeyboard +50% buttons
- Swipe gestures for navigation
- Optimize QR decoding (960x960)
- Joystick for focus direction
- Bindings and bindings!

Apps

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Familiar app life cycle

```
hello.py* 1 from mpos import Activity
           2
           3 class Hello(Activity):
           4
           5     def onCreate(self):
           6         screen = lv.obj()
           7         label = lv.label(screen)
           8         label.set_text('Hello World!')
           9         label.center()
          10         self.setContentView(screen)
```

- Create Screen
- Add Label
- Set Label Text
- Center Label
- Activate Screen

Familiar MANIFEST.JSON

```
1 {  
2 "name": "HelloWorld",  
3 "publisher": "MicroPythonOS",  
4 "short_description": "Minimal app",  
5 "long_description": "Demonstrates the simplest app.",  
6 "full_name": "com.micropythonos.helloworld",  
7 "version": "0.0.2",  
8 "category": "development",  
9 "activities": [  
10     {  
11         "entrypoint": "assets/hello.py",  
12         "classname": "Hello",  
13         "intent_filters": [  
14             {  
15                 "action": "main",  
16                 "category": "launcher"  
17             }  
18         ]  
19     }  
20 ]  
21 }
```

- Name
- Publisher
- Descriptions
- Version
- Entrypoints

Familiar file structure

```
apps/com.micropythonos.helloworld/  
├── assets  
│   └── hello.py  
├── META-INF  
│   └── MANIFEST.JSON  
└── res  
    ├── mipmap-mdpi  
    │   └── icon_64x64.png
```

- Code (hello.py)
- MANIFEST.JSON
- Icon(s)
- Zipped to .mpk
- See docs

AppStore backends

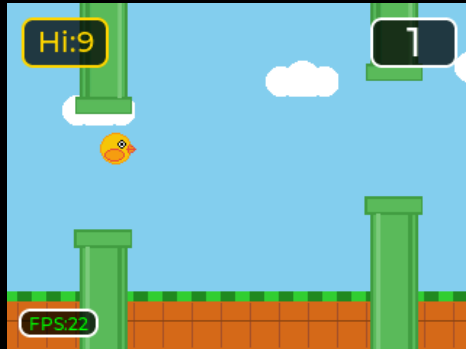
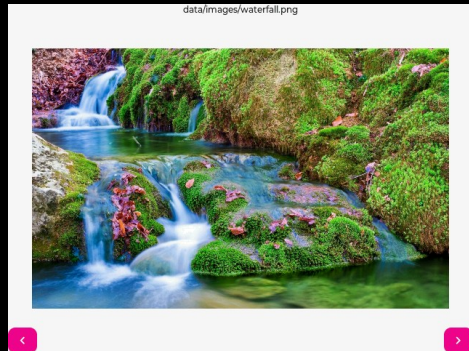
The screenshot shows the BadgeHub web interface for submitting an app. The top navigation bar includes 'Browse Projects', 'Create Project', 'My Projects', 'API Docs', and a user profile for 'Thomas Farstrike'. The main form is divided into several sections:

- Basic Information:** Includes fields for 'App Name' (ImageView), 'Author' (MicroPythonOS), 'Git URL' (https://github.com/MicroPythonOS/MicroPythonOS/tree/main/internal_filesystem/apps/com.micropythonos.imageview), 'Version' (0.0.6), and a 'Description' (Image viewer that opens and shows images on the display. Allows deletion as well.). There is also a 'Hidden' checkbox.
- Categorization:** Features three columns for selection: 'Badge' (with a search bar and a list of badges including 'fr3d_2024'), 'Category' (with a search bar and a list of categories including 'Graphics'), and 'License Type' (set to 'MIT').
- Files:** An 'Upload Files' section with a 'Browse...' button and a note: 'No files selected. You can upload only the type (e.g., code, images, docs). Executable file types will be selectable as "None".'
- Files (Project Files):** A table listing the project files for 'com.micropythonos.imageview_0.0.6.epk':

File Name	Size	Type	Action
com.micropythonos.imageview_0.0.6.epk	17.08KB	None	None
com.micropythonos.imageview_0.0.6_64x64.png	3.18KB	64x64px	Set as icon
icon-64x64.png	2.50KB	64x64px	Icon
metadata.json	0.47KB		

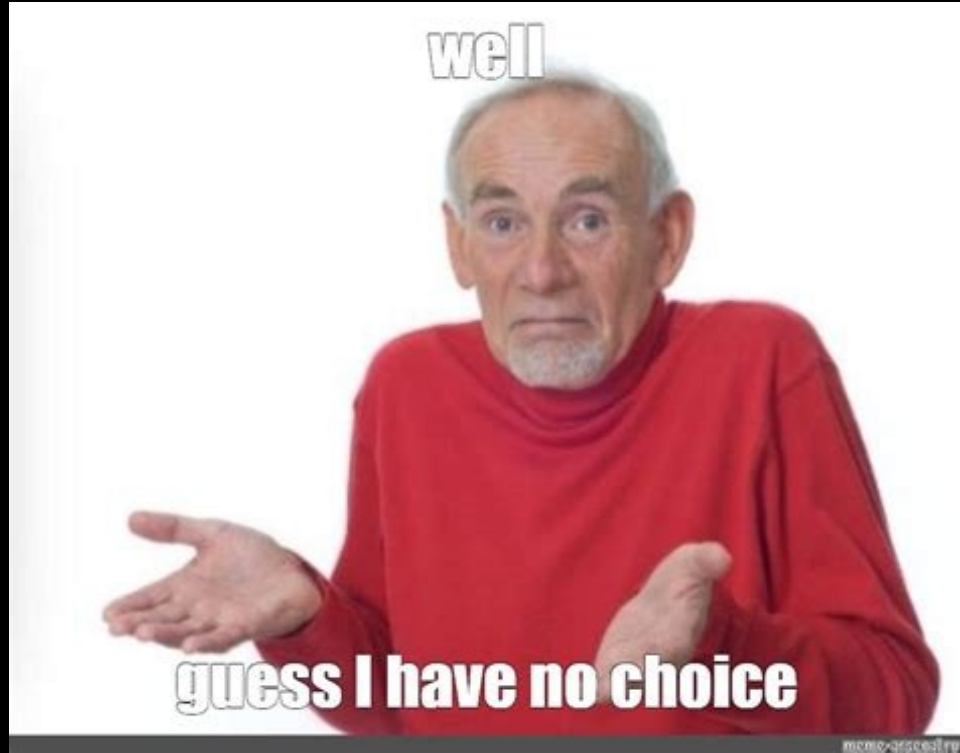
- apps.MicroPythonOS.com
 - Original
 - Curated List
 - To be listed: reach out
- New: BadgeHub.eu
- Add your own!

Apps in the AppStore



- LightningPiggy
- Camera (with QR)
- Image Viewer
- Games
- Music Player
- Sound Recorder
- + 8 more

Why build a MicroPythonOS app



- Fun
- Easy (or Hard)
- Educational
- Many OSS examples
- Supportive community
- $1/1 > 1/100$
- Growing niche

Good documentation

MicroPythonOS Documentation

MicroPythonOS

Search

GitHub

0.7.1 319 20

MicroPythonOS Documentation

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SettingActivity

SettingsActivity

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WidgetAnimator

WifiService

Architecture

MicroPythonOS Documentation

Welcome to the official documentation for **MicroPythonOS**, a lightweight and versatile operating system built entirely in MicroPython. Designed for microcontrollers like the ESP32 and desktop systems, MicroPythonOS offers a modern, Android-inspired interface with an app ecosystem, App Store, and Over-The-Air (OTA) updates.

To see MicroPythonOS in action, check out the screenshots and screen videos on the [main website](#).

About MicroPythonOS

MicroPythonOS is tailored for developers and innovators building IoT devices, smart wearables, educational tools, and more. Its key features include:

- **Native MicroPython:** Write apps and drivers in MicroPython for rapid development.
- **Cross-Platform:** Runs on ESP32 microcontrollers, Linux desktops, and potentially MacOS and Raspberry Pi.
- **Modern UI:** Android-like touch screen interface with gestures.
- **App Ecosystem:** Built-in apps and an App Store for easy expansion.
- **OTA Updates:** Seamless system and app updates.

This documentation provides everything you need to get started, understand the architecture, build the system, and develop apps.

Who This Is For

- **Developers:** Learn how to install, build, and extend MicroPythonOS with new apps and features.
- **Managers:** Explore the system's capabilities, supported hardware, and ready-to-use features for prototyping and production.

Getting Started

New to MicroPythonOS? Start [here](#).

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About MicroPythonOS

Who This Is For

Getting Started

Contributing

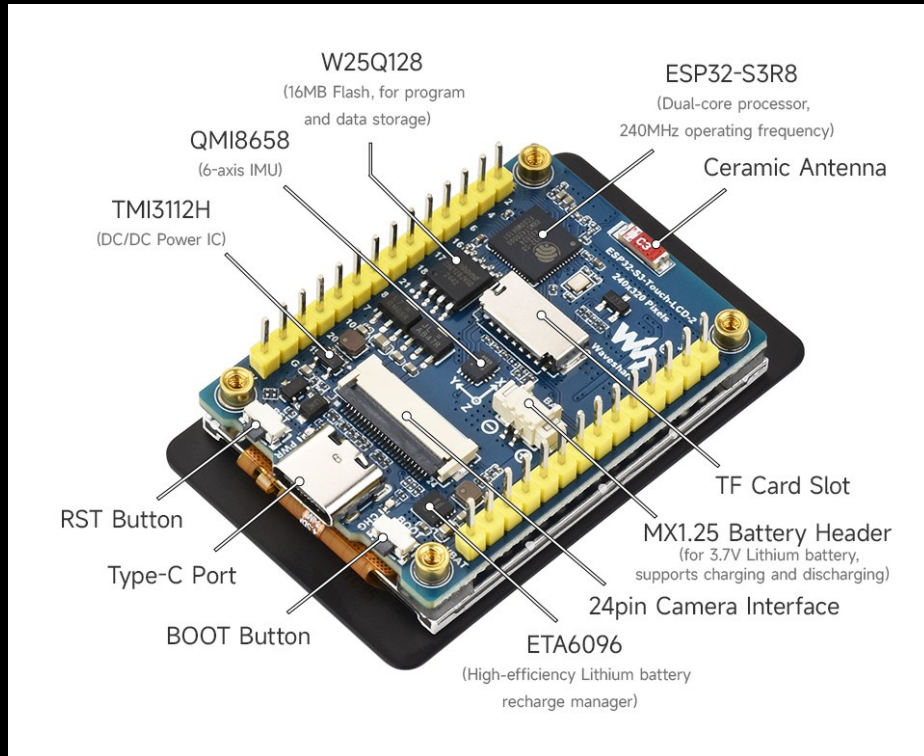
- All frameworks documented
- Updated regularly

<https://docs.MicroPythonOS.com>

Supported hardware

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First board: Waveshare S3 Touch 2



- 320x240
- 2 inch screen
- Capacitive Touch
- Camera 5MP
- Accelerometer (IMU)
- MicroSD Card
- Battery support

Hardware Specs



- 300KB SDRAM
- 8MB PSRAM
- 16MB ROM

Hardware Specs



- 300KB SDRAM
- 8MB PSRAM
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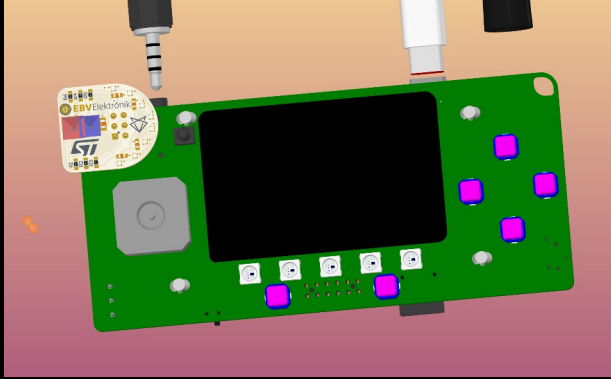
=~ IBM PS/2 (1981)

Second board: Fri3d Camp 2024



- 296x240
- 2 inch screen
- IMU
- Buzzer
- Multicolor LEDs
- Expansions

Third board: Fri3d Camp 2026



- 320x240 2 inch display
- Capacitive Touch!
- Audio Jack
- MicroPythonOS
- Retro-Go
 - Doom
 - Gameboy
 - NES

More boards every months

 Waveshare ESP32-S3-Touch-LCD-2 Fully Supported Learn More	 M5Stack Fire Fully Supported Learn More	 ODROID-Go Fully Supported Learn More	 Matouch ESP32-S3 SPI IPS 2.8 with Camera OV3660 Fully Supported Learn More
 Fri3d Camp 2024 Badge Fully Supported Learn More	 Lilygo T-Display S3 Fully Supported Learn More	 Lilygo T-Watch S3 Supported like T-Watch S3 Plus	 Lilygo T-Watch S3 Plus Supported but some parts (LoRa, GPS) coming soon

- Manufacturers are interested – adds value
- 1st: samples received
- 2nd: samples shipped
- New form factors
- LoRa
- Porting for free
 - Hurry if you want it

Live Demo



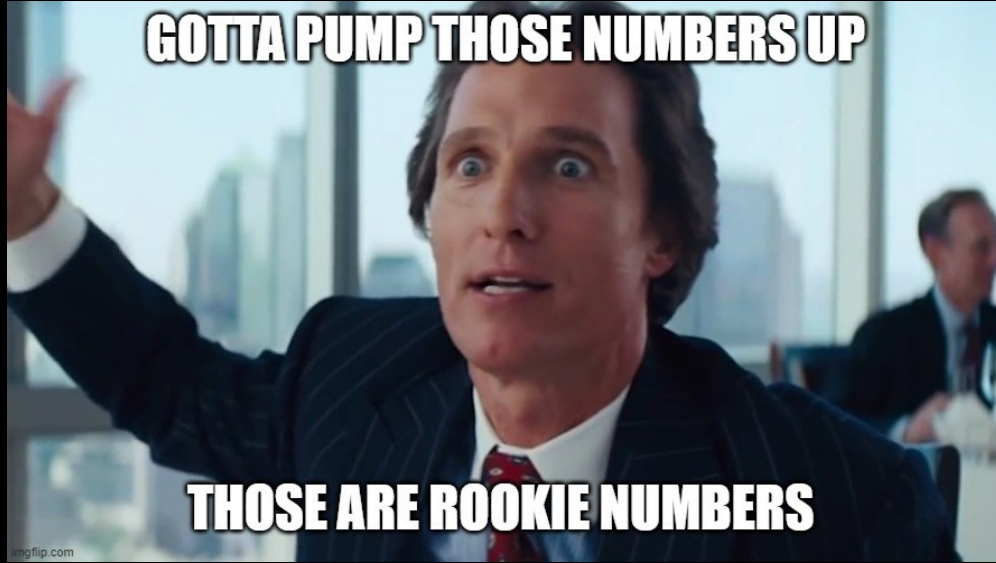
- Demo's are risky
- Demo's are scary
- Will it work?
- Where is it?

Live Demo



- You're looking at it!
- You've been it...
the whole time!

MicroPythonOS in Numbers



- Core Contributors: 14
- Watching: 13
- Apps: 15
- Forks: 71
- Community Chat: 170
- GitHub stars: 715
- Code commits: 2900

Roadmap



- Factory installs
- More frameworks
- API 1.0
- More drivers
- More boards
- Community driven

Next Steps: free stuff



- Run it on desktop
- Prebuilt binaries:
 - Linux
 - MacOS
 - WSL
- Build an app

Next Steps: buy stuff



- Get a supported device
- Flash it
- Install your app
- Publish in AppStore
 - Multiple backends

Easy Installation

Install MicroPythonOS

Click the button below to flash [MicroPythonOS](#) firmware onto your ESP device using WebSerial. Ensure your device is connected via USB and you're using a supported browser (e.g., Chrome or Edge).

Don't have a device yet? Buy a supported one, like the [Waveshare ESP32-S3-Touch-LCD-2](#). Or check out [the docs](#) to see how you can run it on your desktop PC.

Select Version

- Unified for all ESP32-S3 devices: Waveshare and Fri3d Camp 2024
Badge: version 0.7.1

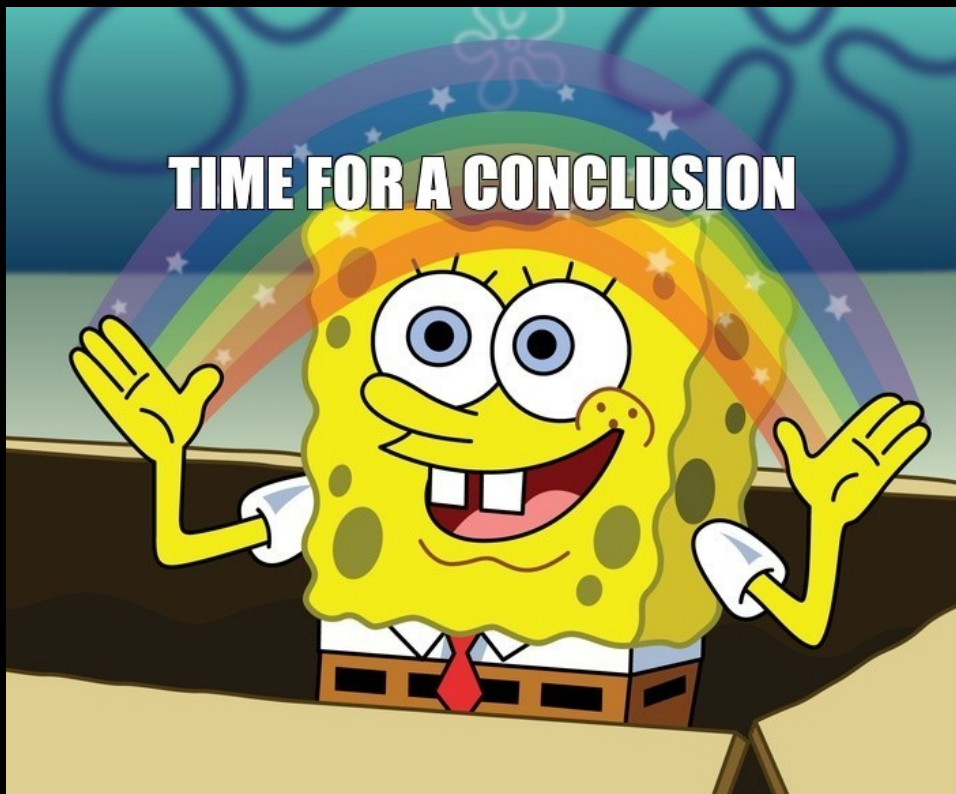
Show old releases ▼

Connect and Install

- WebSerial installer
- One firmware to rule them all

<https://install.MicroPythonOS.com>

Conclusion



- MicroPythonOS is waiting for you
- Join the chat

<https://chat.MicroPythonOS.com>

- Clone the repo
- Questions?

MicroPythonOS

Thank you!
